Notation and Setup for Design-Based Causal Inference*

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This guide collects, in one place, the notation and formal setup shared by the companion guides on (i) the unbiasedness of the Difference-in-Means estimator for both the Sample Average Treatment Effect (SATE) and the Population Average Treatment Effect (PATE), (ii) the variance of the estimator for the SATE and the estimation of that variance, and (iii) the variance of the estimator for the PATE and the estimation of that variance. Each of those guides gives just enough of the setup for its own proofs. This guide develops that setup in full. A reader who is already comfortable with the potential outcomes framework for randomized experiments can safely skim this guide.

1 Units, assignments, and the assignment mechanism

Let the index $i \in \{1, \ldots, n\}$ run over n units in a finite sample, $\mathcal{S}_n$, where $n \geq 4$. Of these n units, the design assigns $n_T \geq 2$ to the treatment condition and $n_C \geq 2$ to the control condition, where $n_T + n_C = n$. The bounds $n_T \geq 2$ and $n_C \geq 2$ (and hence $n \geq 4$) play no role in the unbiasedness of the estimator or in the variance formula itself; they ensure that the conservative (Neyman) estimator of the variance (developed in a companion guide) is well defined. That estimator computes a separate sample variance within each treatment arm, dividing by $n_T - 1$ for the treated units and by $n_C - 1$ for the control units, and a sample variance requires at least two observations; the estimator therefore needs at least two units in each arm, i.e., $n_T \geq 2$ and $n_C \geq 2$, so that $n = n_T + n_C \geq 4$.

Let the binary indicator variable $Z_i \in \{0, 1\}$ record whether unit i receives treatment ($Z_i = 1$) or control ($Z_i = 0$), and collect these indicators in the random assignment vector $\boldsymbol{Z} = \left[Z_1, \ldots, Z_n\right]^\top$. The set of all logically possible assignment vectors is $\{0, 1\}^n$.

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Every assignment vector determines how many units it sends to treatment: Write $n_T(\boldsymbol{z}) \coloneqq \sum_{i=1}^{n} z_i$ for this count, and $n_C(\boldsymbol{z}) \coloneqq n - n_T(\boldsymbol{z})$. Evaluated at the random assignment vector, this counting function yields $n_T(\boldsymbol{Z})$, itself a random variable and a function of $\boldsymbol{Z}$ in exactly the way the potential outcomes below are functions of $\boldsymbol{z}$. The assignment mechanism places positive probability only on a subset of the 2^n logically possible vectors. Under *complete random assignment* the design fixes the count, restricting the support to

$$\Omega = \left\{ \boldsymbol{z} \in \{0, 1\}^n : n_T(\boldsymbol{z}) = n_T \right\},$$

where n_T is a constant that the experimenter chooses.

On Ω the random count $n_T(\boldsymbol{Z})$ equals the constant n_T with probability one, so I write the plain constant, just as z_i denotes the realized value of Z_i. The set Ω is the support of $\boldsymbol{Z}$. Under complete random assignment, $\boldsymbol{Z}$ is distributed uniformly on Ω, and the number of elements in Ω, denoted $|\Omega|$,[1] is $\binom{n}{n_T}$. By contrast, n independent Bernoulli assignments would place positive probability on all 2^n vectors.

I distinguish three objects that are easily conflated. The *assignment space* $\{0, 1\}^n$ is the set of all conceivable assignments. The *support* $\Omega \subseteq \{0, 1\}^n$ is the (typically smaller) set of assignments the mechanism can actually produce. And the *randomization distribution* is the probability distribution the mechanism places on Ω (uniform, under complete random assignment). All of the randomness in what follows comes from this distribution. One convention holds throughout the guides: Ω always denotes the support of $\boldsymbol{Z}$ under the assignment mechanism in force, and when an argument moves between mechanisms I condition explicitly rather than change the symbol. The same way of building a support, holding a count fixed and collecting every vector that achieves it, extends to stratified designs: A blocked or matched study fixes a treated count within each stratum, and the study's support collects the assignment vectors that hit all of those counts at once.

The same framework accommodates *simple random assignment*, under which the mechanism assigns each of the n units independently. Two features change. First, no design constraint pins down the count, so $n_T(\boldsymbol{Z})$ is a nondegenerate random variable. Second, I exclude the two degenerate assignment vectors (all treated and all control), under which one group is empty and the Difference-in-Means estimator is undefined; the support is therefore $\Omega = \{\boldsymbol{z} \in \{0, 1\}^n : 0 < n_T(\boldsymbol{z}) < n\}$, which has $2^n - 2$ elements. The exclusion changes the distribution, so I state the distribution explicitly: $\boldsymbol{Z}$ follows the distribution of n independent Bernoulli draws *conditioned on* the event $\{0 < n_T(\boldsymbol{Z}) < n\}$, and the conditioning breaks the independence of the coordinates. Even though the design does not fix $n_T(\boldsymbol{Z})$, we can fix the count by conditioning on its realized value: The randomization distribution conditional on the event $\{n_T(\boldsymbol{Z}) = n_T\}$ is exactly the complete random

[1] For an arbitrary set W, let $|W|$ denote the cardinality of (i.e., the number of elements in) the set W.

assignment distribution above (Pashley et al., 2021). Hence results that I derive under complete random assignment carry over to simple random assignment after conditioning on the realized number of treated units.

2 The potential outcomes schedule and the Stable Unit Treatment Value Assumption

Adopting the terminology of Freedman (2009) and later Gerber and Green (2012), define a potential outcomes schedule as a vector-valued function $\boldsymbol{y} : \{0,1\}^n \to \mathbb{R}^n$ that maps each possible assignment vector $\boldsymbol{z} \in \{0,1\}^n$ to an n-dimensional vector of real-valued outcomes. The schedule lists how each of the n study participants would respond to every assignment $\boldsymbol{z}$ that the experiment could in principle produce. The schedule is a *fixed* feature of the units; randomness enters only through which assignment the mechanism could select. I move in numbered steps from this general object to the familiar pair of potential outcomes: for each unit, one fixed outcome under treatment and another under control. I will use the same format, general object followed by assumption followed by licensed simplification, whenever an assumption changes what the notation must express.

Step 1 (general object). Unit i's outcome under assignment $\boldsymbol{z}$ is the ith coordinate of the schedule's value, written $y_i(\boldsymbol{z})$. Composed with the random assignment, $y_i(\boldsymbol{Z})$ is a random variable that could in principle take as many as 2^n distinct values, one for each assignment vector.

Step 2 (assumption invoked). Under the Stable Unit Treatment Value Assumption (SUTVA)[2] (Cox, 1958; Rubin, 1980, 1986), unit i's outcome depends on the assignment vector only through unit i's own coordinate: For any $\boldsymbol{z}, \boldsymbol{z}'$ with $z_i = z_i'$, $y_i(\boldsymbol{z}) = y_i(\boldsymbol{z}')$.

Step 3 (licensed simplification). By Step 2, the function $\boldsymbol{z} \mapsto y_i(\boldsymbol{z})$ depends on $\boldsymbol{z}$ only through z_i, so $y_i(1)$ is well defined as the common value of $y_i(\boldsymbol{z})$ across all $\boldsymbol{z}$ with $z_i = 1$, and $y_i(0)$ as the common value across all $\boldsymbol{z}$ with $z_i = 0$. SUTVA thus collapses 2^n numbers per unit to two fixed numbers per unit. The individual causal effect for unit i on the additive scale is $\tau_i \coloneqq y_i(1) - y_i(0)$. The vectors $\boldsymbol{y}(1)$ and $\boldsymbol{y}(0)$ collect the treatment and control potential outcomes for all n units, and $\boldsymbol{\tau}$ collects the n individual effects; all of these are fixed, finite population quantities.

Step 4 (realization). The observable outcome for unit $i \in \{1, \dots, n\}$ is

$$Y_i \coloneqq y_i(Z_i) = Z_i y_i(1) + (1 - Z_i)\, y_i(0),$$

[2] SUTVA implies that (1) units in the experiment respond to only the treatment condition to which each unit is individually assigned and (2) the treatment condition is actually the same treatment for all units assigned to treatment and the control condition is the same for all units assigned to control.

which is random only through Z_i; under the realized assignment $\boldsymbol{Z} = \boldsymbol{z}$, the observed outcome is the number $y_i(z_i)$. Potential outcomes are fixed, the assignment is random, and observable outcomes are random only because a random assignment selects which fixed potential outcome we see.

3 Estimands and the Difference-in-Means estimator

Write $\bar{y}(1) \coloneqq n^{-1} \sum_{i=1}^{n} y_i(1)$ and $\bar{y}(0) \coloneqq n^{-1} \sum_{i=1}^{n} y_i(0)$ for the finite population means of the treatment and control potential outcomes; both are fixed quantities. The target of interest is the Sample Average Treatment Effect (SATE),

$$\tau_{\text{SATE}} \coloneqq n^{-1} \sum_{i=1}^{n} \tau_i = n^{-1} \sum_{i=1}^{n} \big(y_i(1) - y_i(0)\big) = \bar{y}(1) - \bar{y}(0),$$

which is likewise a fixed (though unknown) quantity. The Difference-in-Means estimator of τ_{SATE} is

$$\hat{\tau} \coloneqq \frac{\sum\limits_{i=1}^{n} Z_i Y_i}{\sum\limits_{i=1}^{n} Z_i} - \frac{\sum\limits_{i=1}^{n} (1 - Z_i)\, Y_i}{\sum\limits_{i=1}^{n} (1 - Z_i)}.$$

Unlike the estimand τ_{SATE}, the estimator $\hat{\tau}$ is a *random variable*, since $\hat{\tau}$ is a function of the random assignment vector $\boldsymbol{Z}$ and inherits all randomness from $\boldsymbol{Z}$. The denominators of the ratio form are the counts $n_T(\boldsymbol{Z})$ and $n_C(\boldsymbol{Z})$; under complete random assignment the design fixes them at n_T and n_C, so we can equivalently write $\hat{\tau}$ as

$$\hat{\tau} = \frac{1}{n_T} \sum_{i=1}^{n} Z_i Y_i - \frac{1}{n_C} \sum_{i=1}^{n} (1 - Z_i)\, Y_i.$$

I use the ratio form when the number of treated units may be random (e.g., under simple random assignment) and the form with fixed denominators when n_T and n_C are fixed.

Because the only randomness is the assignment, I write $\mathrm{E}_\Omega[\cdot]$ and $\mathrm{Var}_\Omega[\cdot]$ for expectations and variances over the distribution of $\boldsymbol{Z}$ on Ω, to emphasize that they pertain to the randomization distribution alone. The subscript names the distribution in force on Ω, not the set itself. In these guides that distribution is the uniform one just described. The same support can carry other distributions, however, as when a sensitivity analysis considers departures from uniform assignment.

4 Finite population variances of the potential outcomes

Two equivalent families of finite population variances appear in the variance results. The first uses the divisor $n-1$:

$$S_n^2(\boldsymbol{y}(1)) = \left(\frac{1}{n-1}\right) \sum_{i=1}^{n} \big(y_i(1) - \bar{y}(1)\big)^2$$
$$S_n^2(\boldsymbol{y}(0)) = \left(\frac{1}{n-1}\right) \sum_{i=1}^{n} \big(y_i(0) - \bar{y}(0)\big)^2$$
$$S_n^2(\boldsymbol{\tau}) = \left(\frac{1}{n-1}\right) \sum_{i=1}^{n} \left(\tau_i - \tau_{\mathrm{SATE}}\right)^2 .$$

The second uses the divisor n and is the form that Gerber and Green (2012, Equation 3.4) uses:

$$\sigma_n^2(\boldsymbol{y}(1)) = \left(\frac{1}{n}\right) \sum_{i=1}^{n} \big(y_i(1) - \bar{y}(1)\big)^2$$
$$\sigma_n^2(\boldsymbol{y}(0)) = \left(\frac{1}{n}\right) \sum_{i=1}^{n} \big(y_i(0) - \bar{y}(0)\big)^2$$
$$\sigma_n(\boldsymbol{y}(0), \boldsymbol{y}(1)) = \left(\frac{1}{n}\right) \sum_{i=1}^{n} \big(y_i(0) - \bar{y}(0)\big) \big(y_i(1) - \bar{y}(1)\big) .$$

The two families are related by $\sigma_n^2(\cdot) = \frac{n-1}{n} S_n^2(\cdot)$, and likewise for the covariance term. All of these are fixed, finite population quantities, and they will be the building blocks of the variance of $\hat{\tau}$. The naming rule deserves one explicit statement: The letter S marks the divisor one less than the size, the letter σ marks the divisor equal to the size itself, the subscript names the index set (n for the sample $\mathcal{S}_n$, N for the superpopulation $\mathcal{P}_N$), and the absence of a subscript marks the infinite superpopulation limit introduced at the end of the next section.

5 Extension to a superpopulation: the Population Average Treatment Effect

For results about the Population Average Treatment Effect (PATE), I embed the finite sample in a superpopulation. Consider a superpopulation $\mathcal{P}_N$ of size $N \geq n \geq 4$, with index $i \in \{1, \ldots, N\}$, and let simple random sampling draw n units from $\mathcal{P}_N$ into the experimental sample $\mathcal{S}_n$, leaving the remaining $N - n$ units unsampled. Among the n sampled units, complete random assignment sends $n_T \geq 2$ to treatment and $n_C \geq 2$ to control, exactly as above.

The binary indicator $R_i \in \{0,1\}$ records whether unit i enters the sample ($R_i = 1$) or stays out of it ($R_i = 0$), and I collect these indicators in $\boldsymbol{R} = \begin{bmatrix} R_1, \ldots, R_N \end{bmatrix}^\top$. Write $n(\boldsymbol{r}) := \sum_{i=1}^N r_i$ for the number of units a sampling vector selects; the support of $\boldsymbol{R}$ is then $\Pi = \{\boldsymbol{r} \in \{0,1\}^N : n(\boldsymbol{r}) = n\}$, which holds a count fixed exactly as Ω does, now at the sampling stage. Simple random sampling fixes $n(\boldsymbol{R}) = n$ with probability one and distributes $\boldsymbol{R}$ uniformly on Π, so I write the plain constant n, exactly as I write n_T under complete random assignment. The estimand is the PATE,

$$\tau_{\text{PATE}} := N^{-1} \sum_{i=1}^N \tau_i,$$

and the corresponding population variances, which follow the naming rule of the previous section (letter S, divisor $N-1$, subscript N for $\mathcal{P}_N$), are

$$\begin{aligned}
S_N^2(\boldsymbol{y}(1)) &= \left(\frac{1}{N-1}\right) \sum_{i=1}^N \left(y_i(1) - \frac{1}{N}\sum_{i=1}^N y_i(1) \right)^2 \\
S_N^2(\boldsymbol{y}(0)) &= \left(\frac{1}{N-1}\right) \sum_{i=1}^N \left(y_i(0) - \frac{1}{N}\sum_{i=1}^N y_i(0) \right)^2 \\
S_N^2(\boldsymbol{\tau}) &= \left(\frac{1}{N-1}\right) \sum_{i=1}^N \left(\tau_i - \tau_{\text{PATE}}\right)^2 .
\end{aligned}$$

The divisor $N-1$ pays off in the PATE variance guide: With this convention, the sample variance $S_n^2(\cdot)$ is *exactly* unbiased for $S_N^2(\cdot)$ under simple random sampling, with no correction factor.

In this setting the Difference-in-Means estimator has *two* sources of randomness, sampling $\{R_i\}_{i=1}^N$ and assignment $\{Z_i\}_{i=1}^n$. I write $\mathrm{E}_\Pi[\cdot]$ and $\mathrm{Var}_\Pi[\cdot]$ for expectations and variances over the sampling distribution, $\mathrm{E}_\Omega[\cdot]$ and $\mathrm{Var}_\Omega[\cdot]$ for those over the assignment distribution (conditional on the sample), and the unsubscripted $\mathrm{E}[\cdot]$ and $\mathrm{Var}[\cdot]$ for those over both. The law of total variance links the overall variance to the stage-specific moments,

$$\mathrm{Var}\left[\hat{\tau}\right] = \mathrm{E}_\Pi\left[\mathrm{Var}_\Omega\left[\hat{\tau}\right]\right] + \mathrm{Var}_\Pi\left[\mathrm{E}_\Omega\left[\hat{\tau}\right]\right],$$

which is the organizing identity for the PATE variance derivation. Finally, as $N \to \infty$ with the population variances bounded, the quantities $S_N^2(\cdot)$ converge to limits that I write $\sigma^2(\cdot)$, with no subscript; these limiting superpopulation variances govern the infinite superpopulation results in the PATE variance guide.

References

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